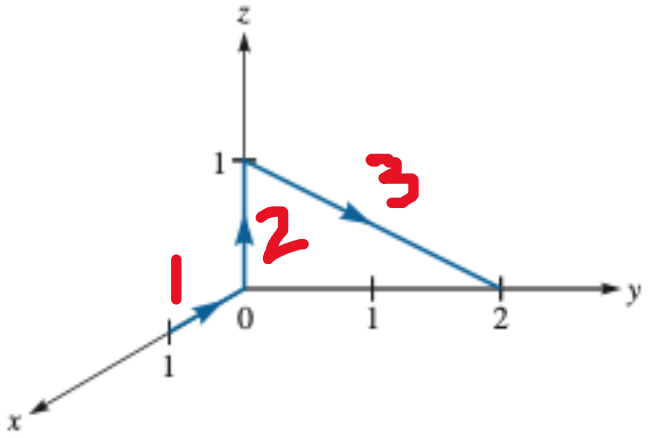
$$\mathbf{H} = (x - y)\mathbf{a}_x + (x^2 + zy)\mathbf{a}_y + 5yz\mathbf{a}_z$$



$$I = \int_\ell \vec{H}^0 \cdot d\vec{\ell} = I_1 + I_2 + I_3$$

$$= \int_{①} \vec{H}^0 \cdot d\vec{\ell} + \int_{②} \vec{H}^0 \cdot d\vec{\ell} + \int_{③} \vec{H}^0 \cdot d\vec{\ell}$$

$$I_1 = \int_{①} H_x\, dx = \int_{x=1}^{0} (x-y)\, dx \Big|_{y=0}$$

$$= \int_{x=1}^{0} x\, dx = \frac{x^2}{2}\Big|_1^0 = -1/2$$

$$I_2 = \int_{②} H_z\, dz = \int_{z=0}^{1} 5yz\, dz \Big|_{y=0} = \int_{z=0}^{1} 0\, dz = 0$$

$$I_3 = \int_{③} (H_y\, dy + H_z\, dz) = \int_{③} \left[(x^2 + zy)\, dy + 5yz\, dz\right]$$

$$= \int_{y=0}^{2} (x^2 + zy)\, dy \Big|_{\substack{x=0 \\ z=-\frac{1}{2}y+1}} + \int_{z=1}^{0} 5yz\, dz \Big|_{y=-2z+2}$$

$$= \int_{y=0}^{2} y\left(1 - \frac{y}{2}\right) dy + \int_{z=1}^{0} 5z(2-2z)\, dz = \frac{2}{3} - \frac{5}{3} = -1$$

$$I = -\frac{1}{2} + 0 - 1 = -\frac{3}{2}$$

$$I_1 = \int_{①} F_x\,dx = \int_{x=2}^{0} x^2 y\,dx\Big|_{y=0} = 0$$

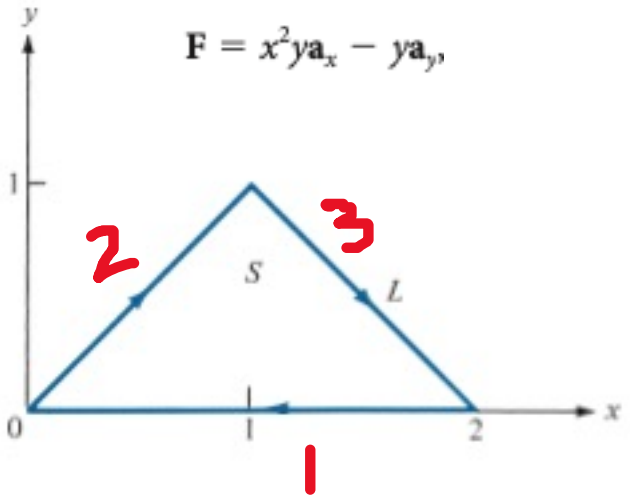
$$I_2 = \int_{②} F_x\,dx + F_y\,dy = \int_{x=0}^{1} x^2 y\,dx\Big|_{y=x} - \int_{y=0}^{1} y\,dy$$

$$= \int_{x=0}^{1} x^3\,dx - \int_{y=0}^{1} y\,dy = \frac{x^4}{4}\Big|_0^1 - \frac{y^2}{2}\Big|_0^1 = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$

$$I_3 = \int_{③} F_x\,dx + F_y\,dy = \int_{x=1}^{2} x^2 y\,dx\Big|_{y=-x+2} - \int_{y=1}^{0} y\,dy$$

$$= \int_{x=1}^{2} x^2(2-x)\,dx - \int_{y=1}^{0} y\,dy = \frac{11}{12} + \frac{1}{2} = \frac{17}{12}$$

$$I_L = I_1 + I_2 + I_3 = 0 - \frac{1}{4} + \frac{17}{12} = \frac{-3}{12} + \frac{17}{12} = \frac{7}{6}$$

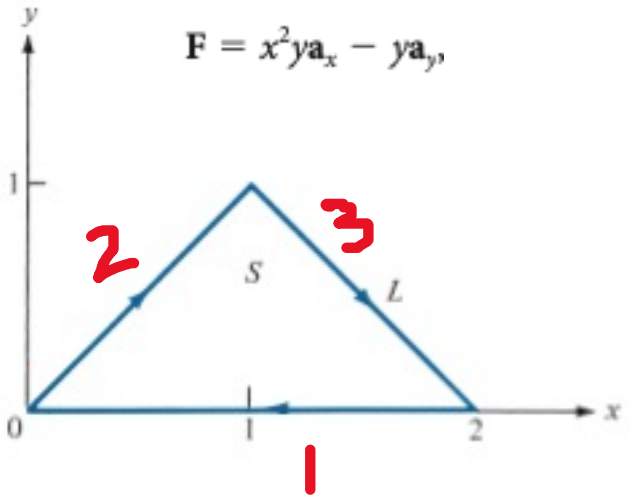


$$\nabla \times \vec{F} = \vec{a}_x \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \vec{a}_y \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) + \vec{a}_z \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

$$= \vec{a}_x (0+0) + \vec{a}_y (0-0) + \vec{a}_z (0-x^2)$$

$$\nabla \times \vec{F} = -x^2 \vec{a}_z$$

$\mathbf{F} = x^2 y \mathbf{a}_x - y \mathbf{a}_y,$

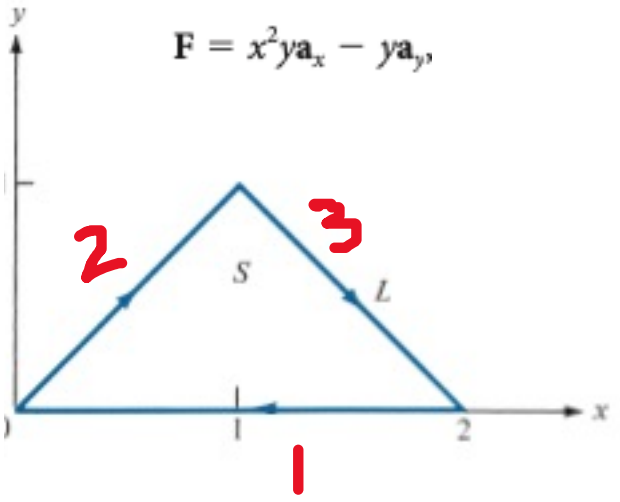


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$$I_S = \int_S (\nabla \times \vec{F}) \cdot d\vec{s} = \int_S (-x^2 \vec{a}_z) \cdot \underbrace{(-\vec{a}_z\, dx\, dy)}_{d\vec{s}}$$

$d\vec{s}$ points in the $-z$ direction in this case.

$$= \int_S x^2\, dx\, dy$$



$\mathbf{F} = x^2 y \mathbf{a}_x - y \mathbf{a}_y,$

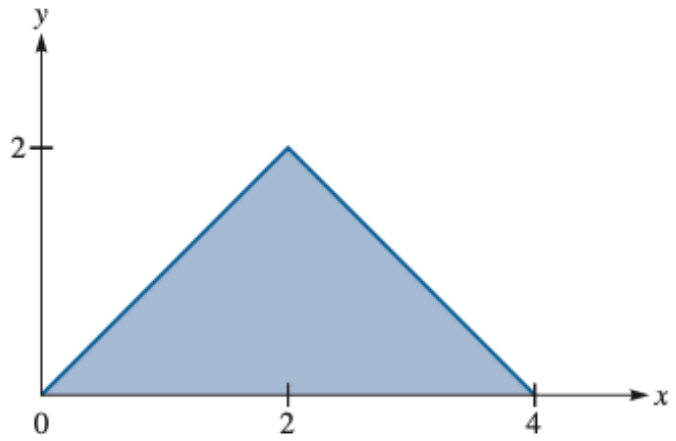
$$= \int_{x=0}^{1} \int_{y=0}^{x} x^2\, dy\, dx + \int_{x=1}^{2} \int_{y=0}^{2-x} x^2\, dy\, dx$$

$$= \int_{x=0}^{1} x^3\, dx + \int_{x=1}^{2} x^2 (2-x)\, dx$$

$$= \frac{1}{4} + \frac{11}{12} = \frac{14}{12} = \frac{7}{6}$$

A cube is defined by $0 < x <$ a, $0 < y <$ a, and $0 < z <$ a. If it is charged with $\rho_v = \frac{\rho_o x}{a}$, where ρ_o is a constant, calculate the total charge in the cube.

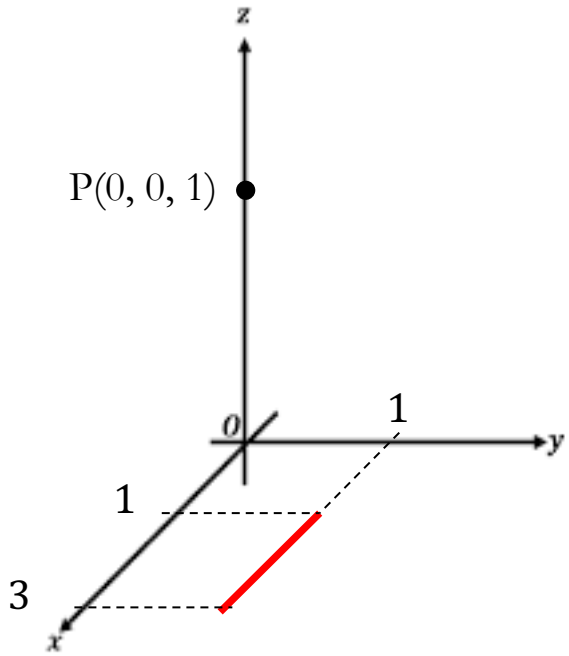
Given that $\rho_s = 6xy\ \mathrm{C/m^2}$, calculate the total charge on the triangular region



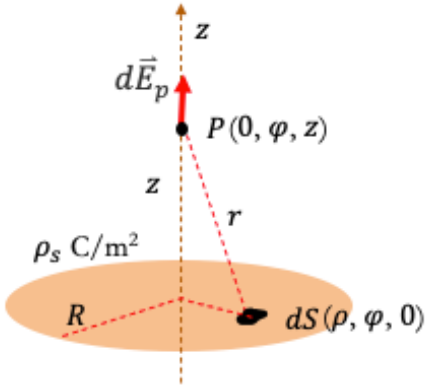
A volume charge with density $\rho_v = 5\rho^2 z$ mC/m^3 exists in a region defined by $0 < \rho < 2$, $0 < z < 1$, $30° < \phi < 90°$. Calculate the total charge in the region.

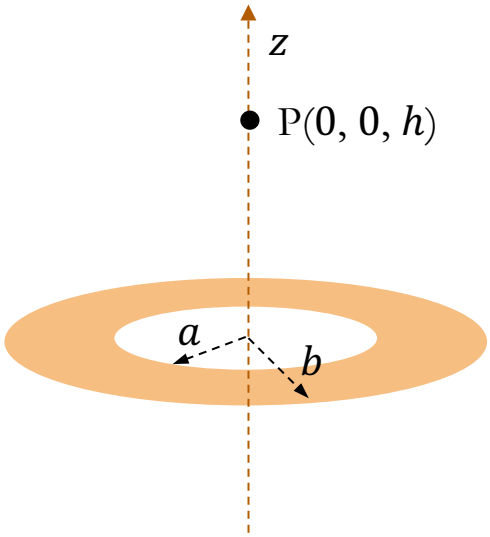
Line $0 < x < 1\,\text{m}$ is charged with density $12x^2$ nC/m. (a) Find the total charge. (b) Determine the electric field intensity at (0, 0, 1000 m).

Find the electric field intensity at point P. Assume uniform line charge density



An annular disk of inner radius a and outer radius b is placed on the xy-plane and centered at the origin. If the disk carries uniform charge with density ρ_s, find **E** at $(0, 0, h)$.





$$\vec{E}_p = \frac{\rho_s}{2\varepsilon}\left[1 - \frac{h}{\sqrt{R^2 + h^2}}\right]\vec{a}_z$$

Only valid along disk axis

$$\vec{E}_p = \frac{\rho_s}{2\varepsilon} h \left[\frac{1}{\sqrt{a^2 + h^2}} - \frac{1}{\sqrt{b^2 + h^2}}\right]\vec{a}_z$$

For annular disk